Logic, Sets and Numbers

Logic

Def. 1. *A **mathematical statement** is an assertion that is strictly true or false. Its negation is $\neg A := A$ is false. A statement depending on variables is a **predicate**.*

Ex. 2.

$$P(x) := x > 2$$

Def. 3. *Quantifiers are short hand notation for making statements about predicates.*

$$forall : \forall x \in \mathbb{R} : P(x)$$

$$exists : \exists x \in \mathbb{R} : P(x)$$

$$exists\ unique : \exists! x \in \mathbb{R} : P(x)$$

$$not\ exists : \nexists x \in \mathbb{R} : P(x)$$

Imp. 4. *The oderder of quantifiers matters. $\forall x \exists y$ is not equivalent to $\exists y \forall x$.*

Def. 5. *Basic logical connectives are defined as:*

$$A \wedge B := A \text{ and } B \text{ are true}$$

$$A \vee B := A \text{ or } B \text{ is true}$$

$$A \implies B := A \text{ implies } B \text{ is true}$$

$$A \iff B := A \implies B \wedge B \implies A$$

Observe that $\neg A \vee B \iff A \implies B$.

Thm. 6. *The following equivalences hold:*

$$\neg(\forall x P(x)) \iff \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \iff \forall x \neg P(x)$$

aswell as

$$\forall x(P(x) \wedge Q(x)) \iff (\forall x P(x)) \wedge (\forall x Q(x))$$

$$\exists x(P(x) \vee Q(x)) \iff (\exists x P(x)) \vee (\exists x Q(x))$$

Sets

Def. 7. *A **set** is a collection of elements. The empty set $\emptyset$ contains no elements. $x \in A$ ($x \notin A$) means x is (not) in A. A is a subset of B denoted by $A \subseteq B$ iff.*

$$\forall x(x \in A \implies x \in B)$$

A is a proper subset of B denoted by $A \subset B$ iff.

$$A \subseteq B \wedge A \neq B$$

Def. 8. *The **union** of A and B is defined as $A \cup B := \{x \mid x \in A \vee x \in B\}$.*

Def. 9. *The **intersection** of A and B is defined as $A \cap B := \{x \mid x \in A \wedge x \in B\}$.*

Def. 10. *The **difference** of A and B is defined as $A \setminus B := \{x \mid x \in A \wedge x \notin B\}$.*

Def. 11. *The **Cartesian product** of A and B is defined as $A \times B := \{(a, b) \mid a \in A \wedge b \in B\}$.*

Numbers

The basic and often used sets of numbers are:

- $\mathbb{N}$: The set of natural numbers 1, 2, 3, 4, ...
- $\mathbb{Z}$: The set of integers ... -2, -1, 0, 1, 2, ...
- $\mathbb{Q}$: The set of rational numbers $\frac{p}{q}$ where $p, q \in \mathbb{Z}$
- $\mathbb{R}$: The set of real numbers

Rem. 12. *Some extensions of the real numbers are*

$$\mathbb{R}_0^+/\mathbb{R}^+ := [0, \infty)/(0, \infty)$$

$$\mathbb{R}_0^-/\mathbb{R}^- := (-\infty, 0]/(-\infty, 0)$$

Def. 13. *Continuous subsets are called **intervalls**. An intervall is called*

Def. 14. ***open** if it does not contain its endpoints*

$$I = (a, b) := \{x \mid a < x < b\}$$

Def. 15. ***closed** if it contains its endpoints*

$$I = [a, b] := \{x \mid a \leq x \leq b\}$$

Def. 16. ***half open** if it contains one endpoint and not the other*

$$I = [a, b) := \{x \mid a \leq x < b\}$$

Def. 17. *An intervall is **bounded** if it is bounded from above and below hence a and b are finite. A bounded and closed intervall is **compact**.*

Def. 18. *An **upper (lower) bound** of a subset $X \subset \mathbb{R}$ is an element $y \in \mathbb{R}$ such that:*

$$\forall x \in X : x \leq (\geq) y$$

Def. 19. *The **maximum (minimum)** of a subset $X \subset \mathbb{R}$ is the element $x_0 \in X$ such that:*

$$\forall x \in X : x \leq (\geq) x_0$$

Def. 20. *Least upper bound and greatest lower bound are also called Supremum and Infimum.*

*The **Supremum** $sup(X)$ of a subset $X \subset \mathbb{R}$ is the least upper bound of X. and can be characterized by:*

$$\forall x \in X : x \leq sup(X) \wedge$$

$$\forall \epsilon > 0 : \exists x \in X : x > sup(X) - \epsilon$$

*The **Infimum** $inf(X)$ of a subset $X \subset \mathbb{R}$ is the greatest lower bound of X. and can be characterized by:*

$$\forall x \in X : x \geq inf(X) \wedge$$

$$\forall \epsilon > 0 : \exists x \in X : x < inf(X) + \epsilon$$

Thm. 21. *The **maximum and minimum are unique** aslong as they exist.*

*Every **non empty, bounded from above (below)** subset of $\mathbb{R}$ has a **maximum (minimum)**.*

Def. 22. *A set X is said to be (order)complete if for every non empty subsets $A, B \subseteq X$*

$$\forall a \in A \forall b \in B : a \leq b$$

then there there exists an element $c \in X$ such that:

$$\forall a \in A : a \leq c \wedge \forall b \in B : c \leq b$$

Thm. 23. *The **Archimedian Principle** says that*

$$\forall x \in \mathbb{R}, \forall y > 0 \exists n \in \mathbb{N} : ny > x$$

$$\forall \epsilon > 0 \exists n \in \mathbb{N} : \frac{1}{n} < \epsilon$$

Thm. 24. *The Young Inequality states that for all $\epsilon > 0$ and all $x, y \in \mathbb{R}$:*

$$2|xy| \leq \epsilon x^2 + \frac{1}{\epsilon} y^2$$

Thm. 25. *$\mathbb{R}$ has order axioms: reflexivity, transitivity, antisymmetry, totality, making it an **ordered and order-complete field.***

Def. 26. *For $x, y \in \mathbb{R}^n$, the standard scalar product is defined as:*

$$x \cdot y = \langle x, y \rangle := \sum_{i=1}^{n} x_i y_i$$

Def. 27. *The Euclidean norm and distance are defined as:*

$$||x|| := \sqrt{\sum_{i=1}^{n} x_i^2}$$

$$d(x, y) := \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Thm. 28. *The triangle inequality holds for all $x, y, z \in \mathbb{R}^n$:*

$$||x - z|| \leq ||x - y|| + ||y - z||$$

Def. 29. *The **complex numbers** $\mathbb{C}$ are numbers of the form:*

$$z = a + ib \quad \text{where } a, b \in \mathbb{R}$$

a is the real part $Re(z)$, and b is the imaginary part $Im(z)$. The imaginary unit i has the property $i^2 = -1$.

Def. 30. *Arithmetic of complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$:*

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$z_1 \cdot z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2)$$

Def. 31. *The complex conjugate of $z = x + iy$ is $\bar{z} = x - iy$. Division is calculated by expanding with the conjugate of the denominator:*

$$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{z \cdot \bar{w}}{|w|^2}$$

Rem. 32. *Some usfull properties of the conjugate:*

- $z \cdot \bar{z} = |z|^2$
- $z + \bar{z} = 2Re(z)$ *and* $z - \bar{z} = 2iIm(z)$
- $\overline{z + w} = \bar{z} + \bar{w}$ *and* $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$

Def. 33. *Geometrically, complex numbers can be represented in the Gaussian plane. The **absolute value** of $z = a + ib$ is its **distance from***

***the origin**:*

$$|z| = \sqrt{a^2 + b^2}$$

The distance between z_1 and z_2 is $d = |z_1 - z_2|$. The triangle inequality holds in $\mathbb{C}$: $|z + w| \leq |z| + |w|$.

Thm. 34. *The Euler's formula states that for $t \in \mathbb{R}$:*

$$e^{it} = \cos(t) + i\sin(t)$$

This allows writing trigonometric functions using complex exponentials:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}, \quad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

Eulers formula leads us to a new representation of complex numbers: the polar-coordinates.

Def. 35. *For $z \in \mathbb{C}$, the polar-coordinates are defined as:*

$$z = r(\cos(t) + i\sin(t)) = re^{it}$$

where $r = |z|$ is the absolute value and $t = \arg(z)$ is the argument.

Multiplication and division in polar-coordinates is much easier than in cartesian coordinates:

$$z_1 \cdot z_2 = r_1 e^{it_1} \cdot r_2 e^{it_2} = r_1 r_2 e^{i(t_1+t_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{it_1}}{r_2 e^{it_2}} = \frac{r_1}{r_2} e^{i(t_1-t_2)}$$

So is exponentiation or taking roots:

$$z^n = (re^{it})^n = r^n e^{int}$$

$$\sqrt[n]{z} = \sqrt[n]{r} e^{i\frac{t+2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

Def. 36. *The n-th roots of unity are the solutions to $z^n = 1$, which are given by:*

$$z_k = e^{i\frac{2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

Thm. 37. *The sum of the roots of unity is always zero.*

$$\sum_{k=0}^{n-1} e^{i\frac{2\pi k}{n}} = 0$$

Thm. 38. *The fundamental theorem of algebra states that every polynomial of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$ (counting multiplicity). These roots appear in konplex-conjugate pairs.*

Sequences and Series

Sequences

Def. 39. *A **sequence** (Folge) $(a_n)_{n\in\mathbb{N}_0}$ is a list of numbers, mathematically a mapping*

$$f : \mathbb{N}_0 \to X$$

It can be given explicitly ($a_n = f(n)$) or recursively ($a_n = g(a_{n-1}, a_{n-2}, \ldots)$).

Def. 40. *A sequence (a_n) is called **convergent** with limit $L \in \mathbb{R}$ if:*

$$\forall \epsilon > 0 \exists N > 0 \text{ s.t. } \forall n > N : |a_n - L| < \epsilon$$

Written as $\lim_{n\to\infty} a_n = L$.

*If no such L exists, the sequence is **divergent**. Sequences that diverge to $\pm\infty$ are a special case of divergent sequences.*

Thm. 41. *The limit of a sequence is unique aslong as it exists.*

Thm. 42. *A subsequence $(a_{n_k})_{k\in\mathbb{N}_0}$ of a convergent sequence converges to the same limit L.*

Def. 43. *An **accumulation point** (Häufungspunkt) $A \in \mathbb{R}$ of a sequence fulfills:*

$$\forall \epsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N \text{ such that } |a_n - A| < \epsilon$$

A is an accumulation point iff there exists a convergent subsequence with limit A.

Thm. 44. *Limit arithmetic: For $\lim a_n = K$ and $\lim b_n = L$:*

- $\lim(a_n \pm b_n) = K \pm L$
- $\lim(a_n \cdot b_n) = K \cdot L$
- $\lim(a_n/b_n) = K/L$ *(if $L \neq 0$ and $b_n \neq 0$)*

Thm. 45. ***Sandwich-Theorem:** If $a_n \leq b_n \leq c_n$ for $n \geq N_0$, and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.*

Rem. 46. *Important limits:*

- $\lim_{n\to\infty} \frac{\ln n}{n} = 0$
- $\lim_{n\to\infty} n^{1/n} = 1$
- $\lim_{n\to\infty} x^{1/n} = 1 \quad (x > 0)$
- $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\lim_{n\to\infty} \frac{x^n}{n!} = 0$

Def. 47. *A sequence is **monotonically increasing (decreasing)** if*

$$a_n \leq (\geq) a_m \quad \forall m > n$$

*and **upper (lower) bounded** if there exists M such that*

$$|a_n| \leq (\geq) M \quad \forall n \in \mathbb{N}_0$$

Thm. 48. ***Bolzano-Weierstrass Theorem:** Every bounded sequence has an accumulation point.*

Thm. 49. ***Monotone Convergence Theorem:** Every bounded monotonic sequence converges to its supremum (infimum).*

Def. 50. *The limit of the supremum is called **limes superior** and writen as:*

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup_{k\geq n} a_k$$

*The limit of the infimum is called **limes inferior** and writen as:*

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf_{k\geq n} a_k$$

Def. 51. *A sequence is a **Cauchy sequence** if*

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that}$$

$$\forall m, n \geq N : |a_n - a_m| < \epsilon$$

Thm. 52. *A sequence of real numbers converges iff it is a Cauchy sequence.*

Series

Def. 53. *A **series** (Reihe) is an infinite sum $\sum_{i=0}^{\infty} a_i$. It converges iff its sequence of partial sums converges.*

Thm. 54. ***Necessary condition for convergence***

$$\lim_{n\to\infty} a_n = 0$$

Rem. 55. ***Geometric series:** $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ iff $|x| < 1$. **Harmonic series:** $\sum_{n=1}^{\infty} \frac{1}{n^s}$ converges for $s > 1$.*

Thm. 56. *For **series with non-negative terms** $a_n \geq 0$, the sequence of partial sums is monotonically increasing. **It converges iff it is bounded.***

Thm. 57. ***Comparison test** (Vergleichskriterium): If $c_n \leq b_n \leq a_n$ for $n \geq N$:*

*If $\sum a_n$ converges, then $\sum b_n$ converges (**Majorantenkriterium**).*

*If $\sum c_n$ diverges, then $\sum b_n$ diverges (**Minorantenkriterium**).*

Def. 58. *A series is **absolutely convergent** if $\sum |a_n|$ converges.*

*A **conditionally convergent** series converges, but not absolutely.*

Thm. 59. ***Riemann's rearrangement theorem:** Let $\sum a_n$ be a conditionally convergent series. Then for any $x \in \mathbb{R}$, there exists a rearrangement of the series that converges to x.*

Thm. 60. ***Leibniz Criterion**: For an alternating series $\sum (-1)^n a_n$ with a monotonically decreasing null sequence (a_n), the series converges.*

Thm. 61. ***Cauchy Criterion:** The series $\sum a_n$ converges iff*

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that}$$

$$\forall m, n \geq N : |\sum_{k=n}^{m} a_k| < \epsilon$$

Thm. 62. ***Root Criterion** (Wurzelkriterium):*

$$\rho = \limsup \sqrt[n]{|a_n|}$$

If $\rho < 1$, abs conv; if $\rho > 1$, div.

Thm. 63. ***Ratio Criterion** (Quotientenkriterium):*

$$\rho = \lim \left| \frac{a_{n+1}}{a_n} \right|$$

If $\rho < 1$, abs conv; if $\rho > 1$, div.

Def. 64. *A **power series** (Potenzreihe) centered at a has the form $\sum_{k=0}^{\infty} c_k (x-a)^k$.*

Thm. 65. *Every power series has a **convergence radius** R, with the **interval of convergence** $(a - R, a + R)$. R is computed by the formula $R = \frac{1}{\limsup |c_n|^{1/n}}$.*

Functions

Def. 66. *A **function** $f : D \to C$ is a map from a domian D to a codomain C s.t. for each $x \in D$ there is exactly one $y \in C$ st. $f(x) = y$.*

Rem. 67. *The input is called the* ***independent variable*** *(argument) while the output is called the* ***dependent variable****.*

Def. 68. *A function f is called* ***injective*** *iff*

$$\forall x_1, x_2 \in A : f(x_1) = f(x_2) \implies x_1 = x_2$$

Def. 69. *A function f is called* ***surjective*** *iff*

$$\forall y \in B \exists x \in A : f(x) = y$$

Def. 70. *A function f is called* ***bijective*** *iff f is injective and surjective.*

Def. 71. *The* ***restriction*** *of f under $D' \subset \mathbb{D}$ is defined as*

$$f\restriction_{D'} : D' \to C, x \mapsto f(x) \quad \forall x \in \mathbb{D}$$

Def. 72. *Let $f : D \to C$ and $g : C \to B$ be functions then*

$$g \circ f : D \to B \quad \textit{with}$$

$$g \circ f(x) = g(f(x)) \quad \forall x \in D$$

is called the ***composition*** *of f and g.*

Def. 73. *The* ***identity function*** *$id_D : D \to D$ is defined as $id_D(x) = x \quad \forall x \in D$.*

Def. 74. *If a function $f : D \to C$ is bijective then there exists a unique function $f^{-1} : C \to D$ such that*

$$f(f^{-1}(y)) = id_C = y \quad \forall y \in C \quad \textit{and}$$

$$f^{-1}(f(x)) = id_D = x \quad \forall x \in D$$

f^{-1} is called the ***inverse function*** *of f.*

Def. 75. *A function f is called* ***bouned*** *if $\exists M \in \mathbb{R}$ such that $|f(x)| \leq (\geq) M \, \forall x \in D$.*

Def. 76. *A function $f : \mathbb{R} \to \mathbb{R}$ is called* ***(strictly) monotonically increasing (decreasing)*** *iff*

$$\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) < (>) f(x_2)$$

Thm. 77. *Every strictly monotonic function is injective.*

Def. 78. *A function $f : \mathbb{R} \to \mathbb{R}$ is called*

- ***even*** *iff $\forall x \in \mathbb{R} : f(-x) = f(x)$*
- ***odd*** *iff $\forall x \in \mathbb{R} : f(-x) = -f(x)$*

Def. 79. *The* ***graph*** *of a function $f : D \to C$ is the set of all ordered pairs $(x, f(x))$ for $x \in D$.*

Def. 80. *A graph is* ***konvex*** *if it lies below the line segment connecting any two points on the graph. Similarly it is called* ***konkav*** *if it lies above the line segment connecting any two points on the graph.*

Limits in the context of functions

Def. 81. *Let $f : \mathbb{D} \to \mathbb{R}$ be a function satisfying $\mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$ for all $\delta > 0$ and $x_0 \in \mathbb{R}$. We say that $\lim_{x \to x_0} f(x) = L$ iff*

$$\forall \varepsilon > 0 \exists \delta > 0 s.t.$$

$$x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \implies |f(x) - L| < \varepsilon$$

The limit L is unique.

Def. 82. *A function $f : \mathbb{D} \to \mathbb{R}$ has in x_0 a* ***right-sided limit*** *$\lim_{x \searrow x_0} f(x) = \lim_{x \to x_0^+} f(x) = L$ iff*

$$\forall \varepsilon > 0 \exists \delta > 0 \quad s.t.$$

$$\forall x \in \mathbb{D}(x_0 < x < x_0 + \delta \implies |f(x) - L| < \varepsilon)$$

left-sided limit *$\lim_{x \nearrow x_0} f(x) = L$ iff*

$$\forall \varepsilon > 0 \exists \delta > 0 \quad s.t.$$

$$\forall x \in \mathbb{D}(x_0 - \delta < x < x_0 \implies |f(x) - L| < \varepsilon)$$

Def. 83. *A function $f : \mathbb{D} \to \mathbb{R}$ has a* ***limit for x approaching infinity*** *$\lim_{x \to \infty} f(x) = L$ iff*

$$\forall \varepsilon > 0 \exists M > 0 \quad s.t.$$

$$\forall x \in \mathbb{D}(x > M \implies |f(x) - L| < \varepsilon)$$

limit for x approaching negative infinity *$\lim_{x \to -\infty} f(x) = L$ iff*

$$\forall \varepsilon > 0 \exists M > 0 \quad s.t.$$

$$\forall x \in \mathbb{D}(x < M \implies |f(x) - L| < \varepsilon)$$

Def. 84. *A function $f : \mathbb{D} \to \mathbb{R}$ has in x_0 a* ***improper limit approaching infinity*** *$\lim_{x \to x_0} f(x) = \infty$ iff*

$$\forall N > 0 \exists \delta > 0 \quad s.t.$$

$$\forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) > N)$$

improper limit approaching negative infinity *$\lim_{x \to x_0} f(x) = -\infty$ iff*

$$\forall N > 0 \exists \delta > 0 \quad s.t.$$

$$\forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) < -N)$$

Thm. 85. *Let $A \in \mathbb{R}$ and assume $\lim_{x \to x_0} f(x) = K (\neq \pm\infty)$ and $\lim_{x \to x_0} g(x) = L (\neq \pm\infty)$.*

- $\lim_{x \to x_0} (f(x) + g(x)) = K + L$
- $\lim_{x \to x_0} (f(x) - g(x)) = K - L$
- $\lim_{x \to x_0} (f(x) g(x)) = KL$
- $\lim_{x \to x_0} (f(x)/g(x)) = K/L$
- $\lim_{x \to x_0} (Af(x)) = AK$

- $\lim_{x \to x_0}(f(x)/g(x)) = K/L$ *if* $L \neq 0$

Continuity

Def. 86. *A function* $f : D \to \mathbb{R}$ *is* ***continous*** *at* x_0 *if*

$$\forall \epsilon > 0 \exists \delta > 0 \text{ s.t. } \forall x \in D$$

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon$$

f *is continuous iff for every sequence* $x_n \to x_0$, $f(x_n) \to f(x_0)$.

Thm. 87. *If* f, g *are continuous, then* $f \pm g$, $f \cdot g$, $c \cdot f$, f/g *(if* $g \neq 0$*), and* $f \circ g$ *are also continuous.*

Thm. 88. *If* $f : I \to J$ *is continuous and bijective,* J *is an interval and* $f^{-1} : J \to I$ *is continuous.*

Thm. 89. ***Intermediate value theorem****: Let* $f : [a, b] \to \mathbb{R}$ *be continuous. For any* $c \in [f(a), f(b)]$*, there is an* $x_0 \in [a, b]$ *with* $f(x_0) = c$.

Def. 90. f *has a* ***local maximum (minimum)*** *at* x_0 *if* $f(x_0) \geq (\leq) f(x) \forall x \in D$*. Both are called* ***extrema****.*

Thm. 91. *A continuous function on a* ***compact interval*** *is bounded and attains its maximum and minimum value.*